

Set up mitmproxy

Privacy Analysis of Augmented Reality Glasses

1 Set Up mitmproxy

- First, on the laptop

1. Get the ip address of the laptop and remember this for a later step.

```
ip a
```

2. Install mitmproxy on the computer.
3. Start mitmproxy using the command

```
mitmproxy -w capture.pcap
```

To have a web-based interface for mitmproxy, one can use the following instead:

```
mitmweb
```

4. Start the hotspot on the computer.

- Secondly, on the phone

1. Connect the phone to said hotspot.
2. After connecting, tap on the right angle to open the network details for the hotspot connection, then swipe down.
3. There should be 'Proxy', tap on it to change the settings to set up the proxy to the following:

Proxy: Manual

Proxy Hostname: ip_address_of_laptop

Proxy Port: 8080 (the default mitmproxy port)

4. Now, open a browser on the phone and visit <http://mitm.it>.
5. Download the suitable certificate for the phone.
6. Install the certificate on the phone, the path could vary depending on the phone model and android version, but the following was the path for me
 - On the old phone:
'Settings', 'Passwords & security', 'Privacy', 'More security settings', 'Encryption & credentials', 'Install a certificate'.
 - On the POCO X5 5G:
'Fingerprints, face data and screen lock', 'Privacy', 'More security settings', 'Encryption & credentials', 'Install a certificate', 'CA certificate'
7. Select CA Certificate and import the downloaded file.
8. If the installation is successful, you should be able to see the credential mitmproxy in 'User credentials'.
9. Now start the app and initiate communication between the glasses and the phone.
10. Observe traffic in mitmproxy or wait for the .pcap file to build.